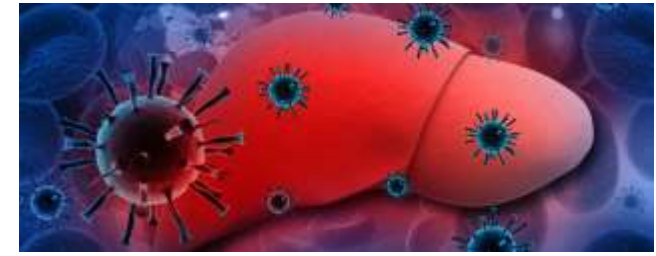


Treatment of hepatitis C



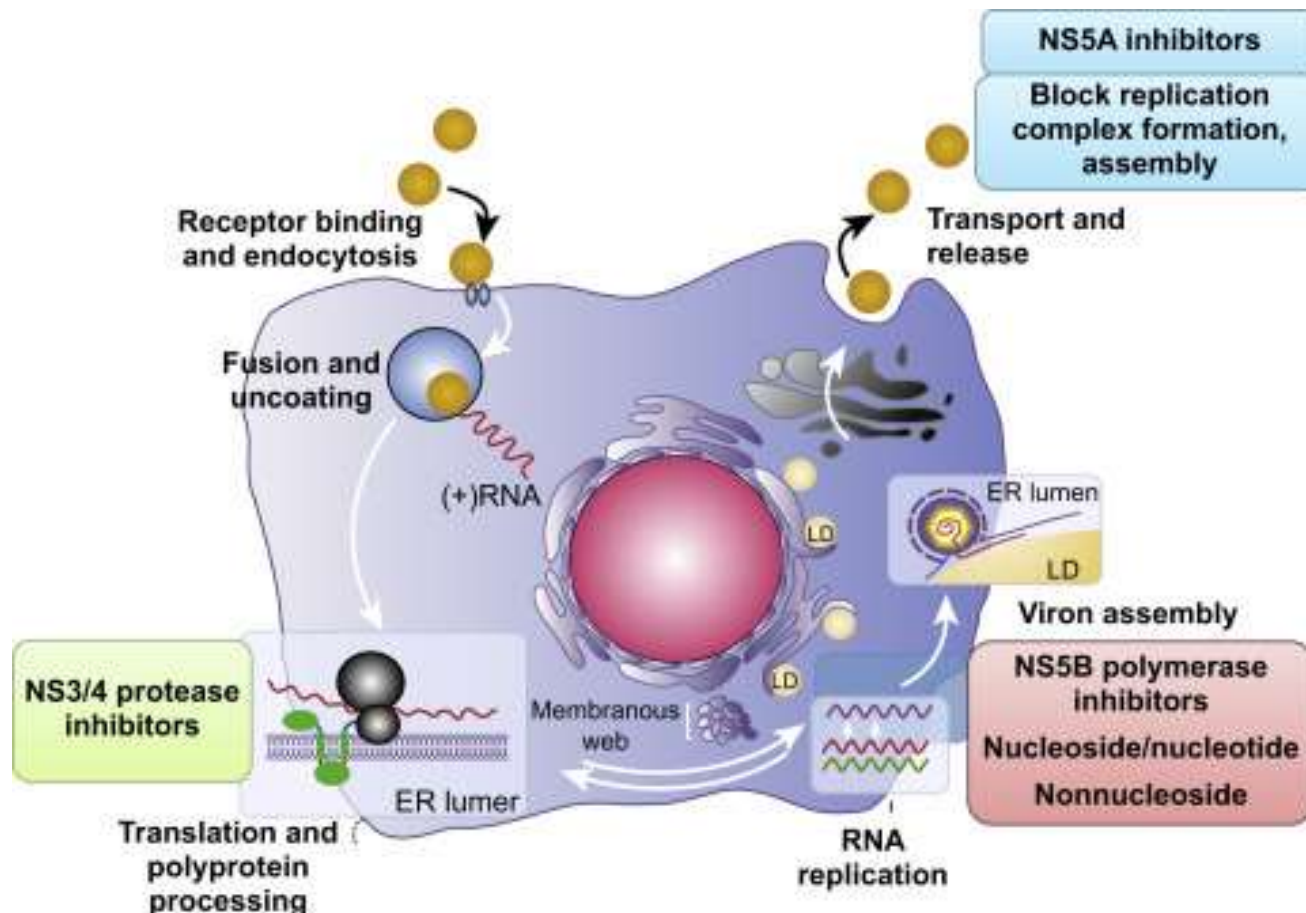
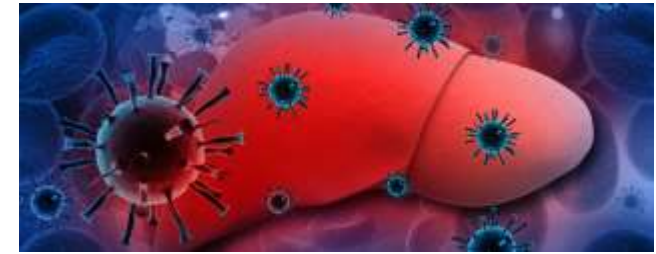
A- Interferon- α -2a:

B- Ribavirin:

- Approved for HCV when **combined** with interferon or with Direct Acting Antivirals (DAAs).
- Combination with DAAs depends on viral genotype/subtype and disease status.
- *Ribavirin* remains an **important** component of HCV therapy, even in the age of DAA therapy.

Treatment of hepatitis C

C- Direct Acting Antivirals (DAAs)

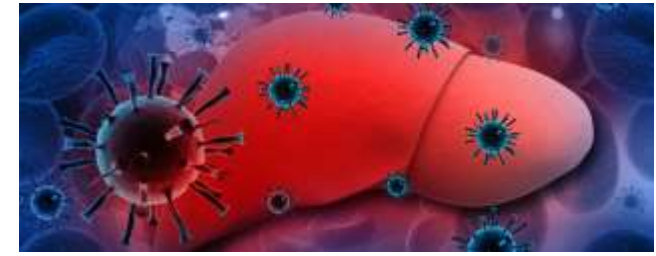


NS3/4A Protease Inhibitors	NS5A Inhibitors	NS5B Polymerase Inhibitors
Boceprevir	Daclatasvir	Dasabuvir
Glecaprevir	Elbasvir	Sofosbuvir
Grazoprevir	Ledipasvir	
Paritaprevir	Ombitasvir	
Simeprevir	Pibrentasvir	
Telaprevir	Velpatasvir	
Voxilaprevir		

Figure 1 - Classes of Direct-Acting Antiviral Agents Used to Treat HCV

Use of **interferon** has become uncommon due to availability of these **DAAs**.

Treatment of hepatitis C



C- Direct Acting Antivirals (DAAs)

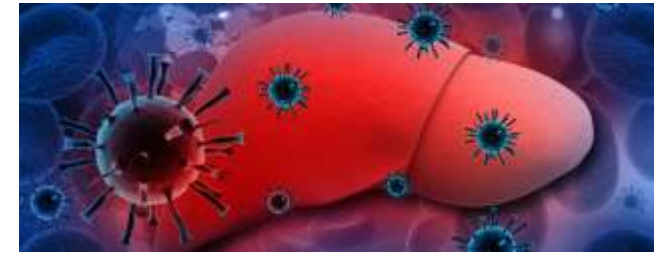
Combinations of DAA



GENERIC NAME(S)	BRAND NAME(S)	APPROVED HCV GENOTYPES
<i>Elbasvir/grazoprevir</i>	Zepatier	1, 4
<i>Glecaprevir/pibrentasvir</i>	Mavyret	1, 2, 3, 4, 5, 6
<i>Paritaprevir/ritonavir/ombitasvir</i>	Technivie	4
<i>Paritaprevir/ritonavir/ombitasvir + dasabuvir</i>	Viekira Pak, Viekira XR	1
<i>Sofosbuvir + daclatasvir</i>	Sovaldi + Daklinza	1, 3
<i>Sofosbuvir/ledipasvir</i>	Harvoni	1, 4, 5, 6
<i>Sofosbuvir/velpatasvir</i>	Epclusa	1, 2, 3, 4, 5, 6
<i>Sofosbuvir/velpatasvir/voxilaprevir</i>	Vosevi	1, 2, 3, 4, 5, 6

Figure 34.7 Combinations of direct-acting antiviral agents for treatment of hepatitis C virus. HCV = hepatitis C virus.

Treatment of hepatitis C



C- Direct Acting Antivirals (DAAs)

A- NS3/NS4A protease inhibitors

- The viral NS3/NS4A proteases are crucial for viral RNA replication.
-  Potential drug-drug interaction due to their inhibition of CYP3A.
- **Adverse effects:** rash, pruritus, nausea, fatigue, and anemia.

NS3/4A Protease Inhibitors

Boceprevir

Glecaprevir

Grazoprevir

Paritaprevir

Simeprevir

Telaprevir

Voxilaprevir

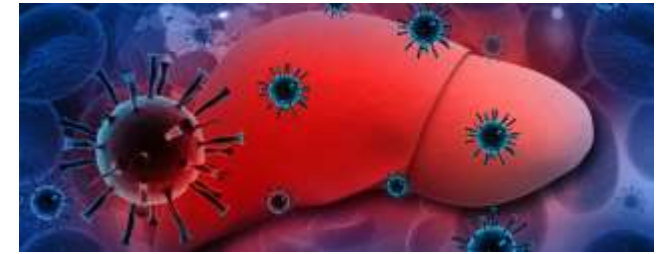
-previr

Treatment of hepatitis C

C- Direct Acting Antivirals (DAAs)

B- NS5A replication complex inhibitors

- NS5A is a viral protein that is essential for HCV RNA replication and assembly.
- Significant drug interactions due to their metabolism by hepatic CYP450 isoenzymes and inhibition of P-glycoprotein (P-gp).



NS5A Inhibitors

Daclatasvir

Elbasvir

Ledipasvir

Ombitasvir

Pibrentasvir

Velpatasvir

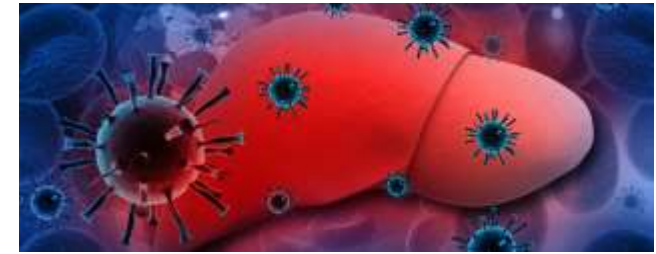
-asvir

Treatment of hepatitis C

C- Direct Acting Antivirals (DAAs)

C- NS5B polymerase inhibitors

- NS5B is the sole RNA polymerase for HCV replication.
- Well-tolerated with few side effects.



NS5B Polymerase Inhibitors

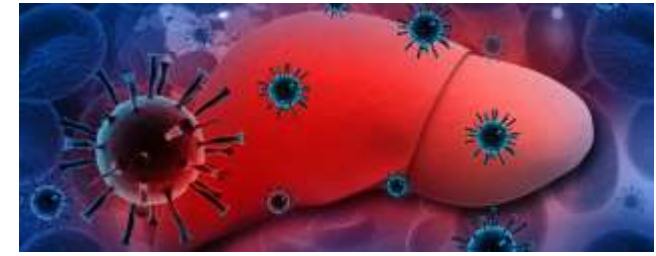
Dasabuvir

Sofosbuvir

-buvir



Treatment of hepatitis C



Latest guideline for the treatment:

- Interferon and ribavirin are no longer recommended.
 - Option 1:
 - Glecaprevir/Pibrentasvir
 - Duration: 8 weeks
 - Option 2:
 - Sofosbuvir/Velpatasvir
 - Duration: 12 weeks
- ✓ These regimens:
- Work against all genotypes
 - Are first-line in current guidelines
 - Cure rates: >95%